Project Report For this activity.

PROJECT REPORT

GPS Navigation System using Graph (C Language)

◆ ABSTRACT

The GPS Navigation System is designed to give another real-world navigation using graph data structures in the C programming language. The system represents locations as nodes and roads as edges with weights indicating distances. It provides functionalities to add, update, delete, and display locations and roads. Additionally, it uses the Dijkstra's Algorithm to give the shortest path between two locations efficiently. This project explains about the practical application of data structures in solving real-world problems like route finding and navigation systems.

◆ INTRODUCTION

In this real-world there is use of digital maps and navigation systems for an efficient route finding. GPS-based systems such as Google Maps work on graph algorithms to determine exact routes. This project mainly used to replace a simplified version of such systems using the C language. By using graphs and shortest path algorithms, the system helps users find the minimum and the maximum distance between two locations.

◆ OBJECTIVES

- To implement graph data structure using C language.
- To perform CRUD operations on locations and roads.
- To calculate the shortest path using Dijkstra's Algorithm.
- To design a menu-driven application.
- To create a real-world GPS navigation functionality

◆ SCOPE OF THE PROJECT

- Supports multiple locations and road connections.
- Allows dynamic memory allocation and deletion of nodes and edges.
- To calculate shortest path between any two locations.

- Works as a basic simulation of navigation systems
- Limitations:
- Does not include real-time traffic updates.
- No graphical user interface (CLI-based only).

🔹 LITERATURE REVIEW

Graph theory is widely used in solving real-world problems like navigation, networking, and logistics. Dijkstra's Algorithm is one of the most operating algorithms for find the shortest path in a graph. Modern navigation systems like Google Maps and Waze use advanced versions of graph algorithms combined with real-time data to provide real routes exactly.

🔹 METHODOLOGY

The project is executed using the following approach:

1. Represent locations as nodes in a graph
2. Represent roads as edges with weights (distance)
3. Use adjacency list for efficient storage
4. Implement CRUD operations for graph management
5. Apply Dijkstra's Algorithm to find shortest path

🔹 TECHNOLOGY USED

- Programming Language: C
- Compiler: GCC
- Data Structure: Graph (Adjacency List)
- Algorithm: Dijkstra's Algorithm
- Platform: Windows.

🔹 Dijkstra Algorithm

Dijkstra's Algorithm is a method used to find the shortest path from a starting node to all other nodes in a weighted graph.

It works by repeatedly selecting the node with the smallest known distance and updating the distances to its neighbouring nodes.

🔹 ARCHITECTURE / SYSTEM DESIGN

The system consists of:

- Node structure → represents locations
- Edge structure → represents roads
- Graph → collection of nodes and edges
- Menu-driven interface → user interaction

Flow:

User Input → Graph Operations → Algorithm Execution → Output Display

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🔹 IMPLEMENTATION DETAILS

The program is divided into modular functions:

- addLocation() → Adds a new location
- addRoad() → Adds a connection between locations
- deleteLocation() → Removes a node
- deleteRoad() → Removes an edge
- updateLocation() → Modifies node data
- displayMap() → Displays graph
- shortestPath() → Implements Dijkstra's Algorithm

Dynamic memory allocation (malloc, free) is used to manage nodes and edges efficiently.

🔹 RESULTS / OUTPUT

The system successfully performs:

- Adding locations and roads
- Displaying graph structure

- Finding shortest path between locations

Example Output:

- Chennai → Vellore (140)
- Vellore → Bangalore (210)

Shortest Path:
Chennai → Vellore → Bangalore
Distance = 350

🔹 CONCLUSION

The project successfully tells us about the implementation of graph data structures and shortest path algorithms in C. It provides a simplified system of a GPS navigation system. The use of Dijkstra's Algorithm ensures efficient route calculation, making the system reliable for pathfinding operations.

🔹 FUTURE WORK(means where it is used frequently).

- Integration of real-time traffic data
- Development of graphical user interface (GUI)
- Use of advanced algorithms like A*
- Integration with maps API
- Support for large-scale datasets

🔹 REFERENCES

1. Thomas H. Cormen – Introduction to Algorithms
2. Data Structures using C – Reema Thareja
3. Dijkstra, E.W. (1959) – A note on shortest path problems
4. Online resources: GeeksforGeeks, TutorialsPoint